Forced vibration | Lab 5

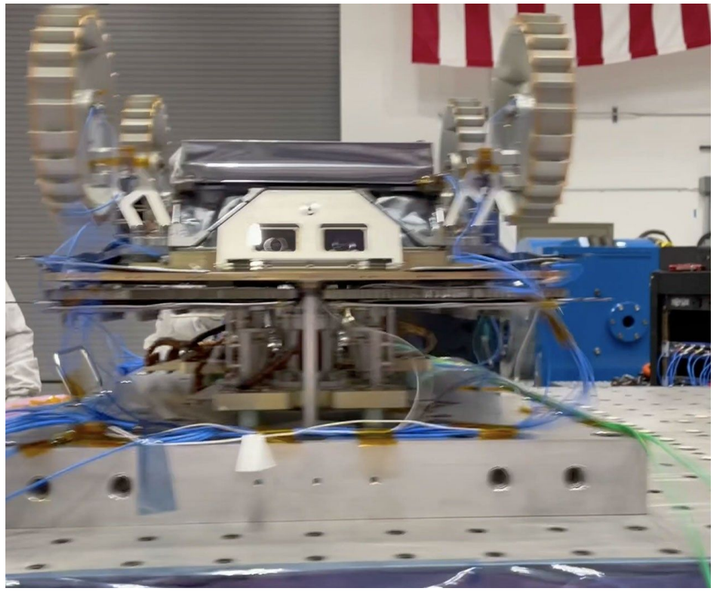
Measure impact decay, periodic response and displacement amplification.



Industrial pump and motor

Rotating forces can excite vibration in the machine and its supports.

Photo: HopuWiki / Wikimedia Commons · CC BY-SA 4.0



Lunar rover shaker test

A shaker imposes motion; engineers measure how the structure responds.

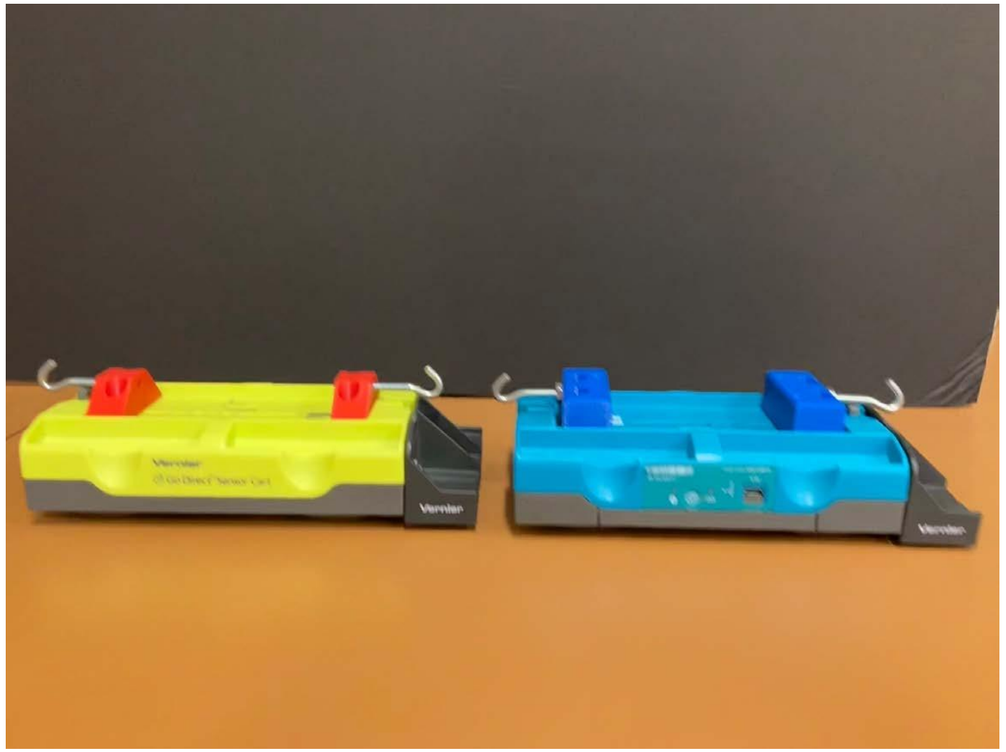
Photo: NASA/JPL-Caltech · CADRE vibration test

Engineering applications

Pump and motor supports: repeated loading can amplify vibration.

Precision stages: vibration limits positioning accuracy and settling.

This cart test identifies which drive speeds cause large motion.

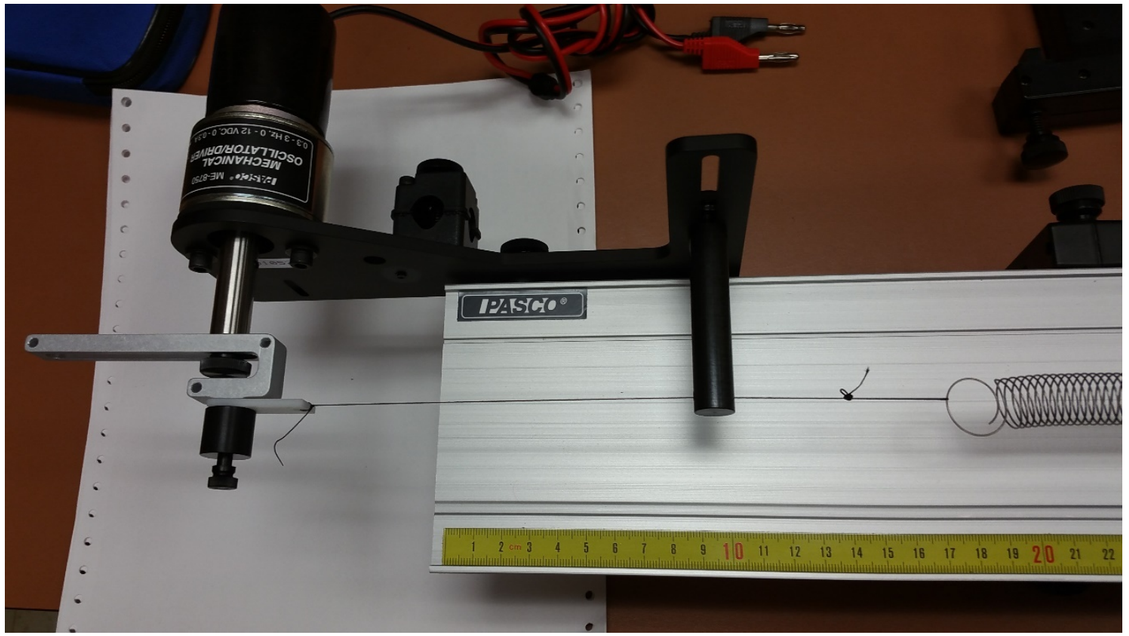


What the apparatus measures

The motor moves one spring end; the cart encoder measures cart position.

Two springs provide restoring force. Magnets provide eddy-current braking.

The tachometer measures drive speed; supply voltage only controls the motor.

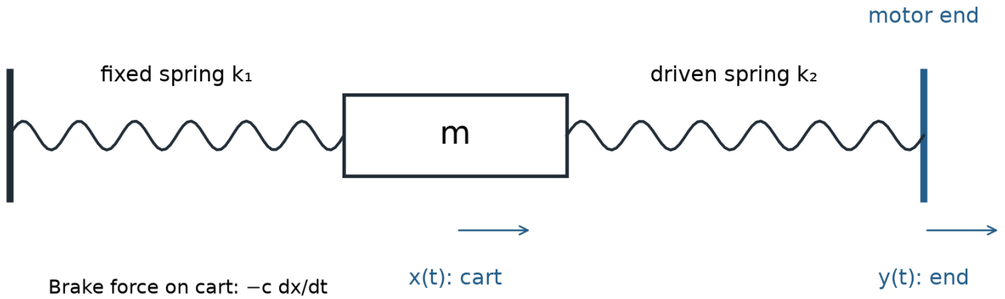


Coordinates and measured parameters

x(t): cart displacement; y(t): driven-end displacement, both positive right [m].

m: total moving mass [kg]; k_1, k_2: spring stiffness [N/m].

c: equivalent braking coefficient [N s/m], acting against the stationary track.

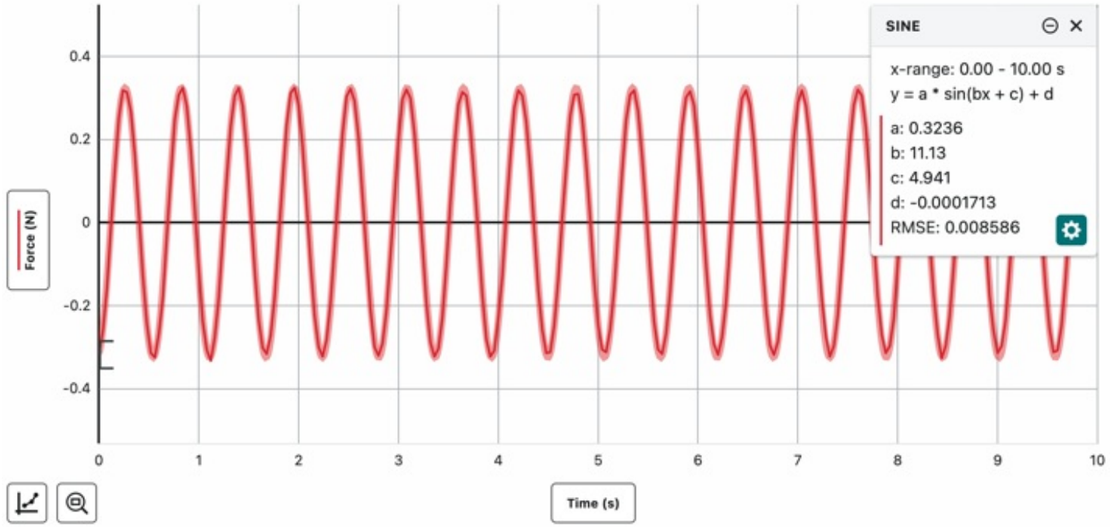


Spring and mass measurements

Weigh the cart, added hex rods and mounted dampers.

For a hanging calibration mass m_h, fit A sin(Bt + phase) + offset.

B is angular frequency [rad/s]; convert Hz before calculating stiffness.



$$k \approx m_h B^2, \qquad B = 2\pi f, \qquad k_{eq} = k_1 + k_2$$

Vernier Graphical Analysis

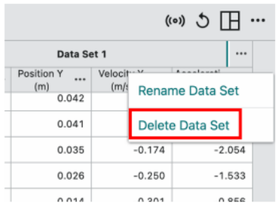
Connect the cart serial number and choose Position versus Time.

Set meters, collection rate at or below 25 Hz, and zero at equilibrium.

Collect before excitation. Export one continuous run as CSV; keep the raw file.







A short tap starts the cart moving

Motor off. Keep both outer spring ends stationary and remove the magnetic dampers. Let the cart rest at equilibrium.

Start recording. Give the cart body one brief, gentle horizontal tap with your fingers, then remove your hand immediately.

Keep the wheels rolling and travel small. Avoid the end stops, spring slack and repeated contact. Analyze after your hand leaves.

The impulse–momentum relation comes from Newton’s law

Let t^- and t^+ bracket the short contact. The cart velocities immediately before and after it are v^- and v^+ [m/s].

Integrate the net force: acceleration is dv/dt, so its time integral is the change in velocity. Constant mass m stays outside the integral.

The net impulse Jnet has units N s = kg m/s. Divide by m only when calculating the velocity change.

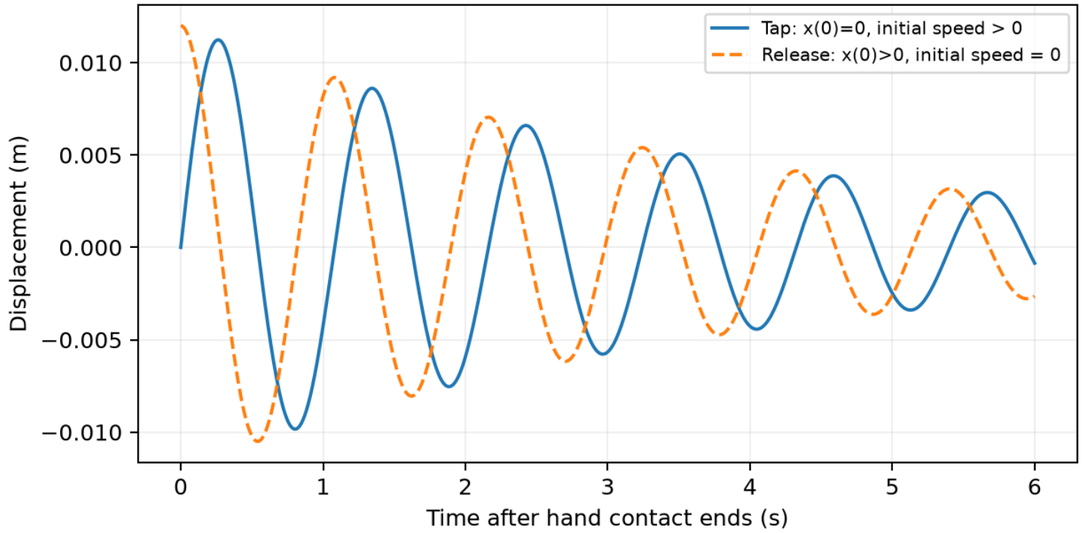
$$J_{\text{net}} = \int_{t^-}^{t^+} F_{\text{net}}\, dt = m \int_{v^-}^{v^+} dv = m(v^+ - v^-)$$

A tap and a pull-and-release set different initial conditions

Tap near equilibrium: position changes little during contact, but the cart gains velocity.

Pull-and-release: hold displacement x_0, then release without pushing. Initial velocity is nearly zero.

Both then follow the same free-decay equation. Initial conditions change the waveform.



Illustrative free decays with the same mass, stiffness and damping

$$\text{tap}: \; x(0) \simeq 0, \; \dot{x}(0^+) \simeq J/m; \qquad \text{release}: \; x(0) = x_0, \; \dot{x}(0) = 0$$

Frequencies and decay rate in the free-response model

Equivalent stiffness is the sum of the two spring stiffnesses. Total moving mass includes every mounted attachment.

The decay rate α [s^{-1}] controls the exponential envelope. Natural and damped angular frequencies have units rad/s.

Subscript n means undamped natural frequency. Subscript d means the frequency of an underdamped free oscillation.

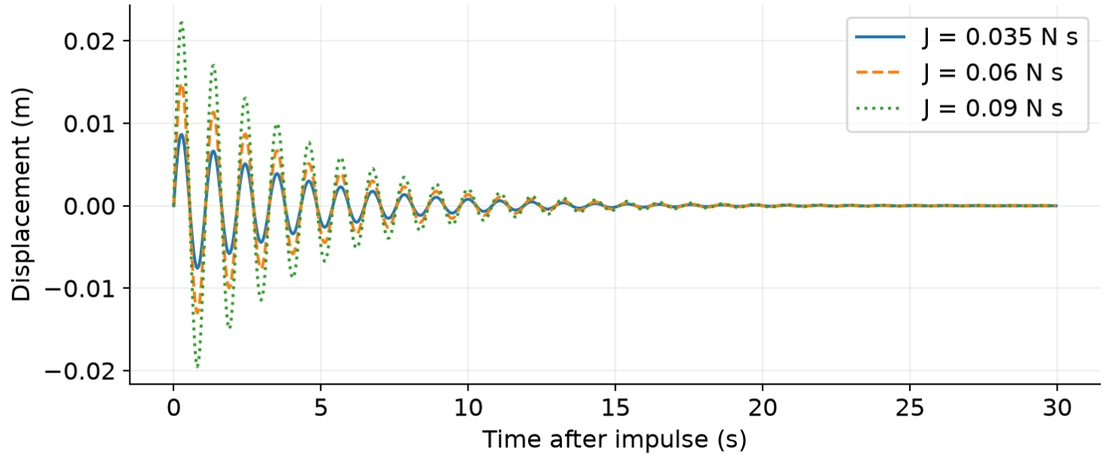
$$k_{eq} = k_1 + k_2, \quad \alpha = \frac{c}{2m}, \quad \omega_n = \sqrt{\frac{k_{eq}}{m}}, \quad \omega_d = \sqrt{\omega_n^2 - \alpha^2}$$

Impulse response follows the free-decay model

For a short tap from equilibrium, the impulse sets initial speed approximately J/m.

The solution starts with nearly zero displacement and the tap-induced velocity.

Larger J increases amplitude. Fixed mass, stiffness and damping give the same frequency and decay rate.



Illustrative model; not laboratory data

$$x(t) = \frac{J}{m\omega_d} e^{-\alpha t} \sin(\omega_d t)$$

Three gentle taps check the effect of excitation level

Record three separate runs: light, medium and slightly stronger gentle taps in the same direction. Start each run from rest near equilibrium.

Keep mass, springs and damper configuration unchanged. Change only the tap level and keep all motion within the linear working range.

Compare displacement amplitudes, fitted frequencies and decay rates. A large frequency change suggests a changed setup or nonlinearity.

How Lab 5 builds on Lab 4

Lab 4: pull and release without pushing. Vary mass, magnetic damping and spring pair to identify their effects on free decay.

Lab 5 impulse tests: keep the setup fixed and vary the brief tap. Check how excitation level affects amplitude and frequency.

Lab 5 motor tests: keep applying a periodic input and vary its frequency. Measure amplification and compare with the forced-response theory.

What an impulse spectrum tells us

The time record reveals period and decay. The FFT is a frequency cross-check.

A finite decaying record spreads spectral energy; do not infer damping from peak height.

The fitted damped frequency and model natural frequency are distinct. Their difference is small for light damping.

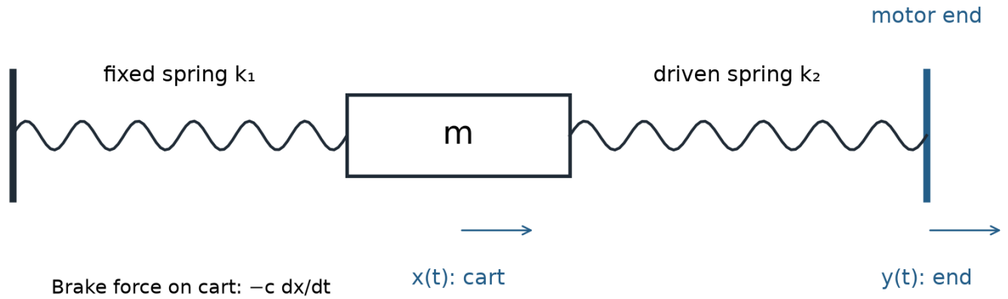
$$f_n = \frac{\omega_n}{2\pi}, \qquad f_d = \frac{\omega_d}{2\pi}$$

Forces from the moving spring end

x(t) is cart displacement. y(t) is the motor-end displacement. Both are horizontal coordinates, positive to the right [m].

The driven spring changes length by y − x, so its force on the cart is $k_2(y - x)$. The fixed spring contributes $-k_1x$.

Sum the two spring forces and the braking force, then collect the terms in x.



$$m\ddot{x} = -k_1x + k_2(y - x) - c\dot{x} \quad \Longrightarrow \quad m\ddot{x} + c\dot{x} + (k_1 + k_2)x = k_2y(t)$$

Why there is no derivative of y in this model

The magnetic brake reacts against the stationary track. Relative brake velocity is therefore the cart velocity.

Only the driven spring connects the cart to the moving endpoint. Its force depends on relative displacement.

A damper between the cart and moving endpoint would add a term proportional to the endpoint velocity. That is a different attachment.

$$F_{\text{brake}} = -c(\dot{x} - 0) \qquad \text{track brake;} \qquad F_{\text{moving damper}} = c(\dot{y} - \dot{x})$$

Prescribed input and settled output

The motor prescribes y(t) = Y cos(ωt). Y is the input peak amplitude [m]; ω = 2πf is drive angular frequency [rad/s].

A linear system in steady harmonic motion responds at the same drive frequency. Its amplitude and phase can differ.

X is cart peak amplitude [m]. ϕ is the output phase lag relative to input [rad]. We will solve for X and ϕ.

$$y(t) = Y\cos(\omega t), \qquad x_{ss}(t) = X\cos(\omega t - \phi)$$

Differentiate the assumed response

The first time derivative shifts cosine to negative sine and multiplies the amplitude by ω.

A second derivative returns cosine with a factor $-\omega^2$. Acceleration therefore opposes displacement.

Insert these derivatives into the force-balance equation from the preceding slides.

$$\dot{x}_{ss} = -\omega X \sin(\omega t - \phi), \qquad \ddot{x}_{ss} = -\omega^2 X \cos(\omega t - \phi)$$

Why we expand the shifted sine and cosine

After substitution, the left side contains shifted functions with phase ϕ, but the input is written as cos(ωt).

Set a = ωt in the identities below. They rewrite both sides using the same two time functions: cos(ωt) and sin(ωt).

This separates the unknown amplitude and phase from time, so we can solve algebraic equations.

$$\cos(a-\phi)=\cos a\cos\phi+\sin a\sin\phi \qquad \sin(a-\phi)=\sin a\cos\phi-\cos a\sin\phi$$

Why both coefficient equations must hold

Define $D = k_1 + k_2 - m\omega^2$ [N/m]. After expansion, the left side is $A\cos(\omega t) + B\sin(\omega t)$.

The right side is $k_2 Y\cos(\omega t) + 0\sin(\omega t)$. Equality must hold throughout every cycle, not just at one instant.

At $\omega t = 0$, A must equal $k_2 Y$. At $\omega t = \pi/2$, B must equal zero. These give the two equations below.

$$DX\cos\phi + c\omega X\sin\phi = k_2 Y, \qquad DX\sin\phi - c\omega X\cos\phi = 0$$

The displacement amplitude depends on input travel

Square the two coefficient equations and add them. The cross terms cancel and $\cos^2\phi + \sin^2\phi = 1$.

The result is $X^2[D^2 + (c\omega)^2] = (k_2Y)^2$. Take the positive square root to obtain the cart peak amplitude X [m].

Y remains in this amplitude formula. At fixed mass, stiffness, damping and drive frequency, doubling input travel doubles X.

$$X = \frac{k_2 Y}{\sqrt{(k_1 + k_2 - m\omega^2)^2 + (c\omega)^2}}$$

A ratio removes the chosen input amplitude

Define gain G = X/Y [dimensionless]. Divide both sides of the amplitude formula by the nonzero input amplitude Y.

Example: if G = 1.5, an input peak travel Y = 10 mm predicts X = 15 mm. With Y = 20 mm, X becomes 30 mm.

The gain stays 1.5 in both cases. The physical motion still depends on Y. This normalization lets us compare different runs.

$$G = \frac{X}{Y} = \frac{k_2}{\sqrt{(k_1 + k_2 - m\omega^2)^2 + (c\omega)^2}}, \qquad X = GY$$

The tangent of phase comes from the sine equation

The zero-sine-coefficient equation is D X sinϕ − cω X cosϕ = 0. Move the damping term to the other side.

For X ≠ 0, D ≠ 0 and cosϕ ≠ 0, division gives tanϕ = cω/D. The inverse tangent therefore helps recover ϕ.

A plain arctangent loses the quadrant. When D < 0, positive damping requires a lag between π/2 and π, not a negative lag.

$$DX\sin\phi = c\omega X\cos\phi \quad \Longrightarrow \quad \tan\phi = \frac{c\omega}{D}, \qquad D = k_1 + k_2 - m\omega^2$$

atan2 selects the physically correct phase lag

Use atan2(cω, D): its two signed arguments retain the quadrant. In MATLAB, atan2 returns radians.

For D > 0 the lag is between 0 and π/2. At D = 0 it is π/2. For D < 0 it lies between π/2 and π.

Example: cω = 1 and D = −1 give ϕ = 135°, whereas atan(−1) gives −45°. Record input and output together to measure this lag.

$$\phi = \text{atan2}(c\omega, k_1 + k_2 - m\omega^2), \qquad x_{ss}(t) = X\cos(\omega t - \phi)$$

Complex amplitudes give the same result

Define $i^2 = -1$. A sinusoid can be written as the real part of a complex amplitude times $\exp(i\omega t)$.

Differentiation then multiplies the complex amplitude by $i\omega$, and a second derivative multiplies it by $-\omega^2$.

The complex ratio H contains both gain and phase: its magnitude is X/Y and its argument is the negative of the lag.

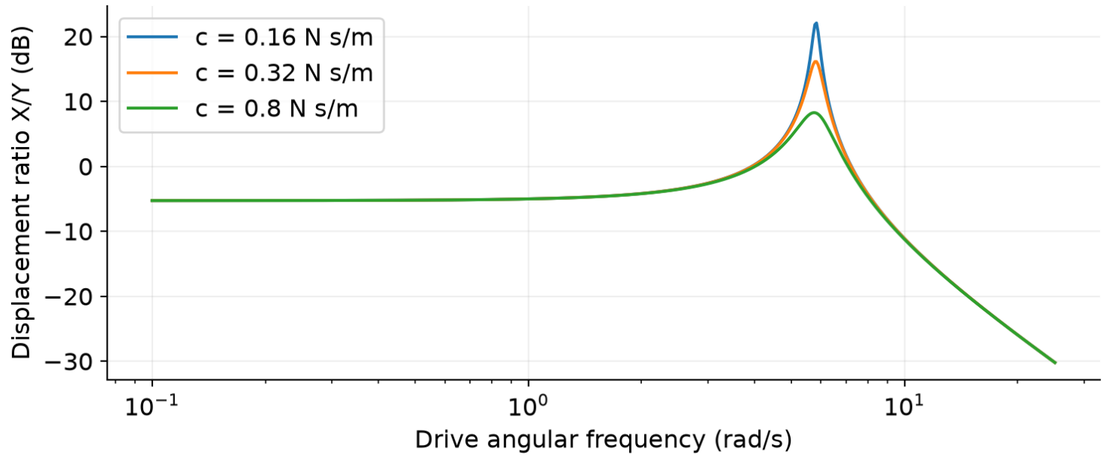
$$H(i\omega) = \frac{\hat{x}}{\hat{y}} = \frac{k_2}{k_1 + k_2 - m\omega^2 + ic\omega}$$

Low-frequency and resonance limits

At low drive frequency, the gain approaches $k_2/(k_1 + k_2)$. Both springs share the endpoint displacement.

Near resonance, inertia partly cancels restoring stiffness. Damping limits the amplitude. The maximum shifts below the natural frequency when damping is nonzero.

ζ is the dimensionless damping ratio. At high drive frequency the cart inertia reduces the displacement ratio.



Illustrative model; not laboratory data

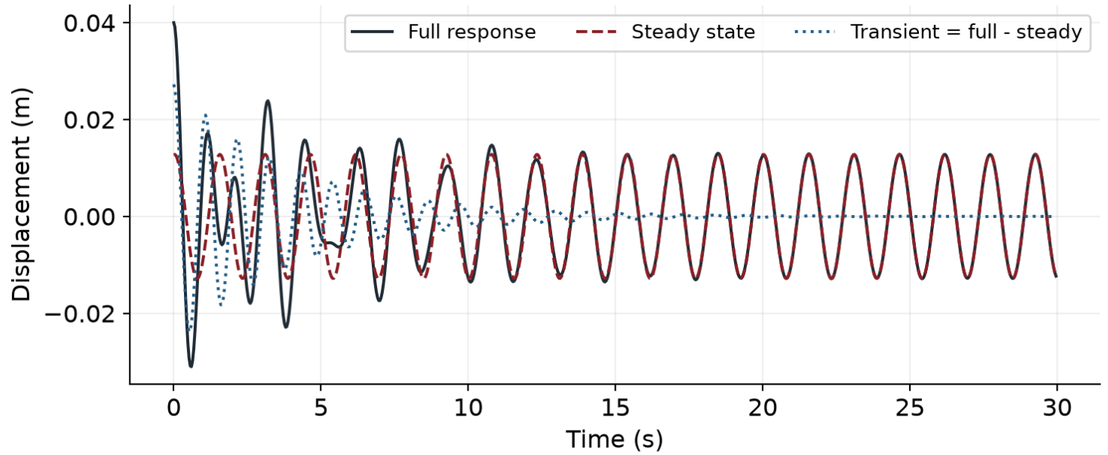
$$\omega_n = \sqrt{\frac{k_1 + k_2}{m}}, \qquad \zeta = \frac{c}{2\sqrt{m(k_1 + k_2)}}$$

The full response includes startup motion

The steady response solves the forced equation; a transient solves the same equation with zero right-hand side.

Their sum satisfies the full equation. The initial position and velocity determine the transient coefficients.

With positive damping, the transient decays and only the forced sinusoid remains. Select that settled interval for the gain comparison.



Illustrative model; not laboratory data

$$x(t) = x_{tr}(t) + x_{ss}(t)$$

How initial conditions determine the transient

Use the decay rate and damped natural frequency defined earlier. The transient contains a decaying sine and cosine.

A and B are transient displacement coefficients [m]. Set them so the total response matches the initial position and velocity.

Changing release conditions changes A and B. It does not change the final harmonic gain for fixed system parameters and drive.

$$x(t) = e^{-\alpha t}[A\cos(\omega_d t) + B\sin(\omega_d t)] + X\cos(\omega t - \phi)$$

Transfer function and state equations agree

A force-input transfer function gives displacement per force [m/N]. Here the effective input force is $k_2 y$.

Multiplying the force-input transfer function by k_2 gives the displacement ratio H [dimensionless].

Set velocity v = dx/dt [m/s]. Dividing Newton's law by m gives the second state equation below.

$$\dot{x} = v, \qquad \dot{v} = -\frac{k_1 + k_2}{m} x - \frac{c}{m} v + \frac{k_2}{m} y$$

Measure input travel and motor speed

Measure actual endpoint peak-to-peak travel, typically 2–3 cm, then divide by 2.

Use the tachometer for each run. One revolution corresponds to one drive cycle.

Record the drive amplitude again when the pin setting or endpoint travel changes.

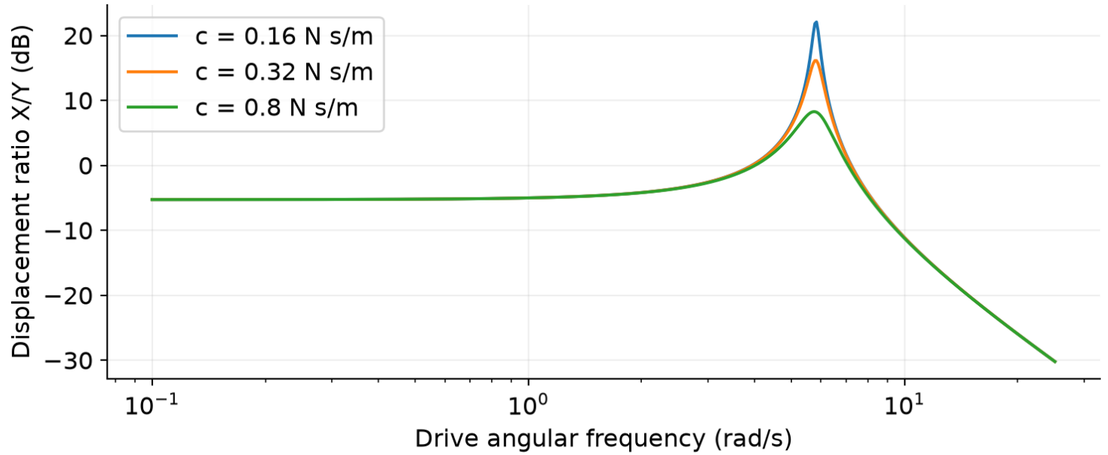
$$Y = \frac{y_{pp}}{2}, \qquad f = \frac{\text{rpm}}{60}, \qquad \omega = 2\pi f$$

A useful sweep covers both sides of resonance

Use 3–5 safe frequencies below and 3–5 above the expected resonance.

Allow startup to decay; record at least five settled cycles, preferably ten.

Keep the springs taut and travel safe. Stop if oscillations become excessive.



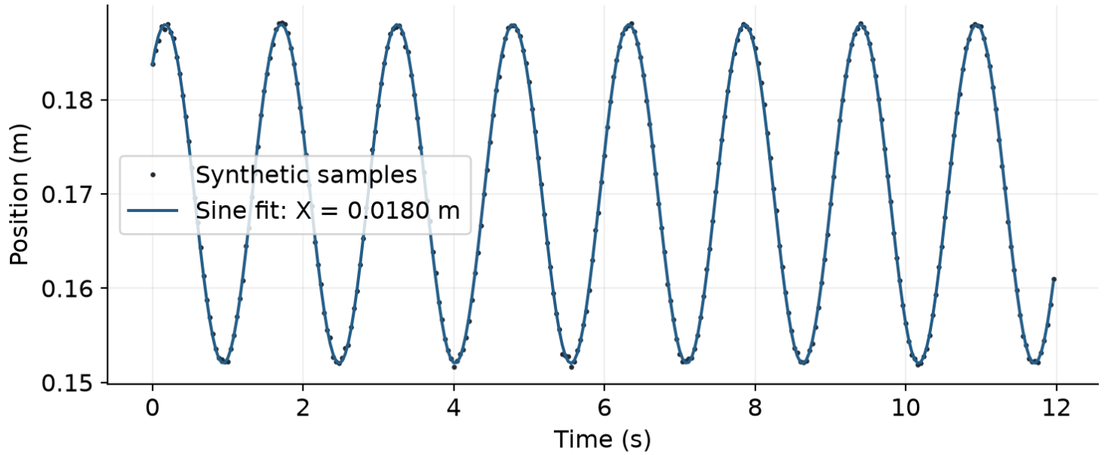
Illustrative prediction used to plan test points

Fit amplitude at the measured drive frequency

Fit $x(t) = \text{offset} + \text{slow drift} + a\cos(\omega t) + b\sin(\omega t)$.

The frequency comes from the tachometer; the code solves for a and b by linear least squares.

The peak amplitude is $X = \sqrt{(a^2 + b^2)}$. Inspect the fit against the data.



Synthetic example of measured-point and fit display

$$G = \frac{X}{Y}, \qquad G_{dB} = 20\log_{10} G$$

Choose damping for the tested configuration

Magnetic dampers change both damping and moving mass.

With the drive off and both ends stationary, record a free decay in the forced-test configuration.

Analyze that decay for c. Use it with the matching mass in the harmonic prediction.

$$c = 2m\alpha, \qquad \omega_d = \sqrt{\frac{k_1 + k_2}{m} - \alpha^2}$$

Test 1: measurements and setup

Weigh the cart, hex rods and all moving attachments. Record the spring IDs and each calibrated stiffness from Lab 4.

Level the track. Keep both springs taut near equilibrium. Set up the stationary ends for impulse runs with magnetic dampers removed.

In Vernier Graphical Analysis, connect the correct cart, select position in meters, set up to 25 Hz and zero at equilibrium. Start collection before each tap.

Test 2: three impulse runs

Let the cart rest at equilibrium. Give one short horizontal tap, remove your hand and record several clean decay cycles.

Repeat from rest at three gentle tap levels without changing the hardware. Save one raw CSV per run.

Use MATLAB mode 1 and select data after contact ends. Compare the three time traces, decay fits and frequency spectra.

Test 3: harmonic-input calibration

Install 1–2 magnetic dampers and reweigh. With both ends stationary and motor off, record a free decay to identify the matching damping coefficient.

With the GTA, measure motor-end peak-to-peak travel at low speed using the guide's calibration setup or a ruler. Input amplitude Y is half that travel.

Stop the motor before reconnecting the driven spring. Measure actual rpm with the tachometer: one revolution gives one input cycle.

$$Y = \frac{y_{pp}}{2}\ [\mathrm{m}], \qquad f = \frac{\mathrm{rpm}}{60}\ [\mathrm{Hz}], \qquad \omega = 2\pi f\ [\mathrm{rad/s}]$$

Test 4: frequency sweep and result checks

With the GTA, increase motor speed gradually. Aim for 3–5 safe frequencies below and 3–5 above predicted resonance. Avoid dwelling at the peak.

At each speed, let the transient decay. Record at least five settled cycles, preferably ten or more. Save raw CSV, rpm and input travel.

Use MATLAB mode 2. Compare measured amplitude and X/Y with theory. Show the GTA one impulse fit, one harmonic fit and the frequency-response points before leaving.

MATLAB workflow

Open Run_Lab5_Forced_Response.m. Enter measured mass, both stiffnesses and matching damping.

Select CSV/Excel files, a settled time interval and position units. Enter measured motor rpm and input peak-to-peak travel.

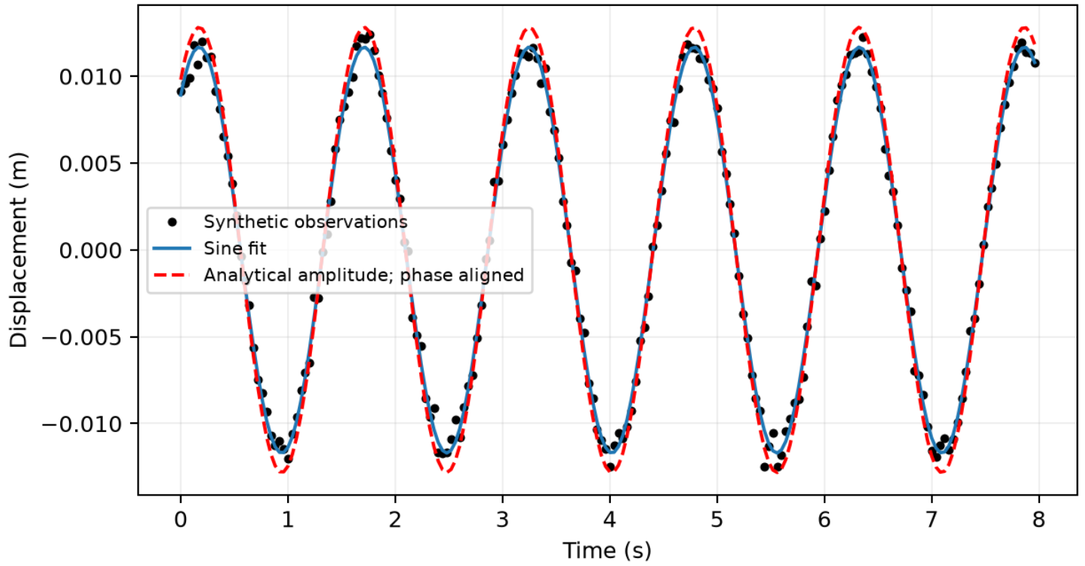
Each run exports measured data, its sine fit and the analytical amplitude comparison. The sweep figure compares measured gain with theory.

What the code compares with theory

The blue sine fit estimates the measured amplitude X. The red dashed curve uses amplitude predicted from m, k_1, k_2, c, Y and drive frequency.

Its phase is aligned to the output for display. That comparison tests amplitude; it does not measure input/output phase.

Read measured X, predicted X and percent difference in the summary. Compare X/Y across drive frequencies.



Synthetic example: fit and analytical amplitude are different quantities

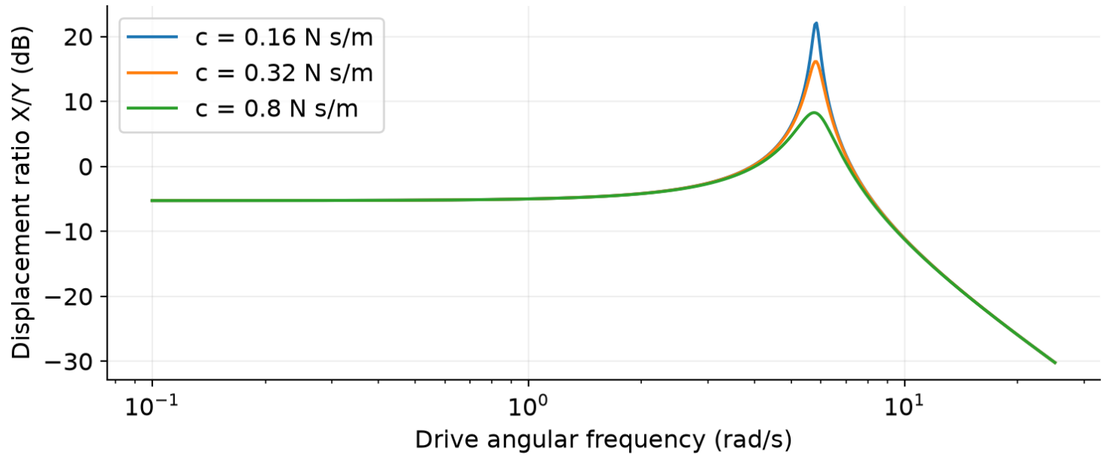
$$X_{\text{theory}} = \frac{k_2 Y}{\sqrt{(k_1 + k_2 - m\omega^2)^2 + (c\omega)^2}}$$

Figures for the combined Labs 4 + 5 report

Three impulse time histories and spectra: compare frequency and amplitude.

Representative harmonic record: identify startup and the settled interval.

Measured displacement ratios overlaid on theory using your measured parameters.



Illustrative theory only; add your own measured points

Check before leaving the laboratory

Show one impulse trace and spectrum, one settled sine fit and at least five sweep points to the GTA.

Confirm time units, meters, kilograms, peak versus peak-to-peak, and rpm conversion.

Copy the raw files and parameter log to every group member.

Three frequencies to distinguish

Natural frequency: set by mass and the two spring stiffnesses.

Damped transient frequency: appears in decay after release or impact.

Drive frequency: remains in the settled forced oscillation.

$$\omega_n \neq \omega_d \neq \omega \quad \text{in general}$$